School of Humanities, Natural & Social Sciences
Department of Mathematics

Course Syllabus

MATH221 - Discrete Mathematics

Spring Semester
2026

Course Title	Discrete Mathematics		
Course Code	MATH221		
Credits / ECTS	6	**Total Hours**	180
Contact hours per week	Lecture / Tutorial		2 / 2
Pre-requisite(s)	MATH101 / None		
Semester	Spring	**Academic Year**	2026
Course leader	Anna Tomskova		
Email	a.tomskova@newuu.uz		
Office Number	115		
Office Hours	Friday, 11am-1pm		
Textbook	Kenneth H. Rosen, Discrete mathematics and its applications, 8th edition, McGraw Hill, 2019		
Reference Books	Susanna S. Epp, Discrete mathematics with applications, 5th edition, Cengage Learning Inc., 2020.		

Course Description (as in the catalog):

This course provides a background in the area of discrete mathematics to provide an adequate foundation for further study in computer science. It is also of great interest to students wishing to pursue further study in mathematics. In this unit, students study propositional and predicate logic; methods of proof; fundamental structures in discrete mathematics such as sets, functions, relations and equivalence relations; Boolean algebra and digital logic; graphs and trees; and elementary counting techniques.

Course Instructors:

Position	Name	Email	Office	Office hours
Professor	Anna Tomskova	a.tomskova@newuu.uz	115	Friday, 11am-1pm

Attendance policy:

Attendance is compulsory. We emphasize the importance of the discussions during the instructed hours. A student missing 25% (32 hours or more) of all instructed hours will be forced to withdraw the course (in accordance with the University Academic Regulations).

Course Learning Outcomes (CLOs):

By the end of successful completion of this course, the student will be able to:

No.	Description
CLO1	Learn and understand propositional logic.
CLO2	Learn, understand, and apply rules of inference, recursive definitions, and proof methods.
CLO3	Learn and understand basic concepts such as sets, functions, sequences, and their properties.
CLO4	Analyze function growth using Big O, Theta, Omega notations.
CLO5	Solve counting problems: permutations, combinations, recurrence relations.
CLO6	Learn and understand relations and elementary graph theory concepts.

Contribution level of Course Learning Outcomes(CLOs) to Student/Program Outcomes

Software Engineering

	SO1	SO2	SO3	SO4	SO5	SO6	SO7
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	Low	High
CLO2	High	Medium	Medium	-	-	Low	High
CLO3	High	Medium	-	-	-	Low	High
CLO4	High	Medium	-	-	-	Low	High
CLO5	High	Medium	Medium	-	-	Low	High
CLO6	High	Medium	Medium	-	-	Low	High

Cyber Security

	SO1	SO2	SO3	SO4	SO5	SO6	SO7
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	Low	High
CLO2	High	Medium	Medium	-	-	Low	High
CLO3	High	Medium	-	-	-	Low	High
CLO4	High	Medium	-	-	-	Low	High
CLO5	High	Medium	Medium	-	-	Low	High
CLO6	High	Medium	Medium	-	-	Low	High

AI & Robotics

	SO1	SO2	SO3	SO4	SO5	SO6	SO7
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	-	-	-	Low	High
CLO2	High	Medium	Medium	-	-	Low	High
CLO3	High	Medium	-	-	-	Low	High
CLO4	High	Medium	-	-	-	Low	High
CLO5	High	Medium	Medium	-	-	Low	High
CLO6	High	Medium	Medium	-	-	Low	High

Applied Mathematics

	SO1	SO2	SO3	SO4	SO5
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration
CLO1	High	High	Medium	-	-
CLO2	High	High	Medium	-	-
CLO3	High	High	-	-	-
CLO4	High	High	Medium	-	-
CLO5	High	High	Medium	-	-
CLO6	High	High	Medium	-	-

Education

	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8
Description	Design and implement effective instructional strategies in STEM disciplines, integrating core disciplinary knowledge with modern pedagogy to enhance student engagement and learning outcomes.	Apply principles of inclusive education, Universal Design for Learning (UDL), and social-emotional learning (SEL) to create equitable and supportive STEM classrooms that address the diverse needs of all students.	Utilize digital tools, including game-based learning and AI technologies, to innovate and improve STEM teaching, learning, and assessment practices.	Develop and evaluate STEM-focused curricula and lesson plans that align with national education standards and contribute to Uzbekistan's strategic goals for innovation and economic growth.	Apply critical thinking and problem-solving skills to address complex challenges in STEM education and foster inquiry-based learning approaches in the classroom.	Conduct research in STEM education, integrating interdisciplinary knowledge and methodologies to improve teaching practices and enhance student outcomes.	Communicate effectively in written, verbal, and digital forms, collaborating with peers, educators, and stakeholders to advance STEM education initiatives.	Demonstrate a commitment to lifelong learning, ethical responsibility, and professional growth through reflective practice and hands-on teaching experiences during the three-year pedagogical internship.
CLO1	-	-	-	-	High	-	Medium	-
CLO2	-	-	-	-	High	Low	Medium	-
CLO3	-	-	-	-	High	-	Medium	-
CLO4	-	-	-	-	High	-	Medium	-
CLO5	-	-	-	-	High	-	Medium	-
CLO6	-	-	-	-	High	-	Medium	-

Weekly Distribution of Course Topics/Contents

Week	Class Type	Topic	Required Reading	Outcome

1	Lecture / Tutorial	Introduction to the course. Propositional Logic: propositions, logical operations, truth tables. Applications of Propositional Logic: translation from logic into language and vice versa, system specifications, logical puzzles.	Chapters 1.1, 1.2, A-11,	CLO1, CLO2, CLO3
2	Lecture / Tutorial	Propositional Equivalences: tautologies, various laws of equivalences. Disjunctive normal form. Predicates and Quantifiers: universal and existential quantifiers, equivalence of statements involving predicates, translations logic vs. language.	Chapter 1.3, 1.4	CLO1, CLO2, CLO3
3	Lecture / Tutorial	Nested quantifiers: equivalence of the nested quantifiers, rules of negation, translation logic vs. language. Independent Study: Rules of Inference: logical syllogisms	Chapters 1.5, 1.6	CLO1, CLO2, CLO3
4	Lecture / Tutorial	Proof Methods and Basic Structures: Sets, Functions, Sequence, Boolean Matrices	Chapter 1.7, 1.8, Chapter 2	CLO1, CLO2, CLO3
5	Lecture / Tutorial	Growth of Functions: Big O, Theta, Omega; Halting Problem	Chapter 3	CLO2, CLO3, CLO4
6	Lecture / Tutorial	Mathematical Induction: mistaken proofs, creative use of induction; Strong Induction: use of strong induction in algorithm correctness	Chapters 5.1, 5.2	CLO2
7	Lecture / Tutorial	Revision Week		CLO1, CLO2, CLO3, CLO4
8	Exam	Midterm Exam		CLO1, CLO2, CLO3, CLO4
9	Lecture / Tutorial	Recursive definitions and Structural Induction: recursively defined functions, Lame Theorem on complexity of the Euclidean algorithm, recursive definition of binary trees Independent study: Program Correctness	Chapters 5.3, 5.4	CLO2, CLO3, CLO5
10	Lecture / Tutorial	Basics of Counting. Generalized Pigeonhole Principle. Permutations and Combinations. Applications of Recurrence Relations. Proof that O(nlogn) comparisons needed to sort an array.	Chapter 6.1, 6.2, 6.3	CLO2, CLO3, CLO5

11	Lecture / Tutorial	Solving Linear Recurrence Relations Recursion revisited. Divide-and-Conquer Algorithms algorithms: modeling using recursion, Master theorem, Karatsuba algorithm of fast multiplication, complexity divide-and-conquer algorithms (Mergesort included), elegant solution of finding closest points on the plane.	Chapters 8.1, 8.2, 8.3	CLO2, CLO3, CLO5
12	Lecture / Tutorial	Relations: types of relations (reflexive, symmetric, transitive), combining relations, composition; Equivalence Relations: equivalence classes and partitions; Partial Orderings: the principle of well-ordered induction, lexicographical order, max and min elements, lattices, topological sort	Chapters 9.1, 9.5, 9.6	CLO2, CLO3, CLO5
13	Lecture / Tutorial	Graph terminology and special types. The handshake lemma, Bipartite graphs, Hall's Marriage Theorem, Hypercube and parallel processing. Graph Representation and Isomorphism: algorithms for graph isomorphisms. Connectivity.	Chapters 10.2, 10.3, 10.4	CLO2, CLO6
14	Lecture / Tutorial	Euler and Hamiltonian Paths: necessary and sufficient conditions, Dirac and Ore's theorems, Gray codes. Planar Graphs: Euler's formula, graph homeomorphisms, Kuratowski's Theorem.	Chapters 10.5, 10.7, 10.8	CLO2, CLO6
15	Lecture / Tutorial	Revision Week		CLO2, CLO3, CLO4, CLO5, CLO6
16	Exam	Final Exam		CLO2, CLO3, CLO4, CLO5, CLO6

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Tutorial activity	weeks 1-7, 9-15	20
Midterm Exam	week 8	40
Final Exam	week 16	40
Total		**100**

Assessment of Course Learning Outcomes:

CLO	Meaning	Evaluation Tool(s) & Weights (tentative)
CLO1	Learn and understand propositional logic.	Tutorial activity (20%) Midterm Exam (80%)
CLO2	Learn, understand, and apply rules of inference, recursive definitions, and proof methods.	Tutorial activity (20%) Midterm Exam (50%) Final Exam (30%)
CLO3	Learn and understand basic concepts such as sets, functions, sequences, and their properties.	Tutorial activity (20%) Midterm Exam (70%) Final Exam (10%)
CLO4	Analyze function growth using Big O, Theta, Omega notations.	Tutorial activity (20%) Midterm Exam (50%) Final Exam (30%)
CLO5	Solve counting problems: permutations, combinations, recurrence relations.	Tutorial activity (20%) Final Exam (80%)
CLO6	Learn and understand relations and elementary graph theory concepts.	Tutorial activity (20%) Final Exam (80%)

Students' Workload:

Students' workload is distributed as follows:

	Activity	Hrs.	%
Assessment methods and criteria	In-Class Time	56	31 %
	Independent Study	55	31 %
	Assignments	51	28 %
	Examinations	18	10 %
	Group Discussions	0	0 %
	Fieldwork and Practicum	0	0 %
	Workshops and Seminars	0	0 %
	Individual Development	0	0 %
Total		**180**	**100%**

Approval			
Course leader - Professor			
Name	**Anna Tomskova**		
Signature		**Date:**	2026-01-25
Department head - Associate Professor			
Name	**Rustam Turdibayev**		
Signature		**Date:**	2026-01-25
Academic Administration - Head of Department			
Name	**Alisher Ikramov**		
Signature		**Date:**	2026-01-26
Department of Quality Assurance and Accreditation - Head of Department			
Name	**Rebecca Lee Carter**		
Signature		**Date:**	2026-01-26